

Lab 03

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```
library(tidyverse)
library(readr)
library(leaflet)
```

1

```
airlines <- read_csv("airlines.csv")
head(airlines)
```

```
# A tibble: 6 x 2
  IATA_CODE AIRLINE
  <chr>      <chr>
1 UA        United Air Lines Inc.
2 AA        American Airlines Inc.
3 US        US Airways Inc.
4 F9        Frontier Airlines Inc.
5 B6        JetBlue Airways
6 00        Skywest Airlines Inc.
```

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
head(airports)
```

```
# A tibble: 6 x 5
  IATA_CODE CITY      STATE LATITUDE LONGITUDE
  <chr>      <chr>      <chr>    <dbl>    <dbl>
1 ABE      Allentown    PA        40.7     -75.4
2 ABI      Abilene      TX        32.4     -99.7
3 ABQ      Albuquerque NM        35.0    -107.
4 ABR      Aberdeen    SD        45.4     -98.4
5 ABY      Albany      GA        31.5     -84.2
6 ACK      Nantucket    MA        41.3     -70.1
```

```
processa_chunk <- function(df, pos) {
  df %>%
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarize(
      soma_atraso = sum(ARRIVAL_DELAY),
      n_voos = n(),
      .groups = "drop"
    )
}

estatisticas_chunks <- read_csv_chunked(
  file = "flights.csv",
  callback = DataFrameCallback$new(processa_chunk),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- estatisticas_chunks %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarize(
    soma_total = sum(soma_atraso),
    n_total = sum(n_voos),
    media_atraso = soma_total / n_total,
    .groups = "drop"
  ) %>%
  select(DESTINATION_AIRPORT, media_atraso)
```

```
head(flights)
```

```
# A tibble: 6 x 2
  DESTINATION_AIRPORT media_atraso
  <chr>                <dbl>
1 ABE                  5.80
2 ABI                  4.08
3 ABQ                  5.61
4 ABR                 -3.39
5 ABY                  8.68
6 ACK                  4.91
```

2

```
aeropertos_ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

aeropertos_ausentes
```

```
# A tibble: 0 x 2
# i 2 variables: DESTINATION_AIRPORT <chr>, media_atraso <dbl>
```

```
flights <- flights %>%
  left_join(
    airports %>% select(IATA_CODE, CITY, STATE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )

head(flights)
```

```
# A tibble: 6 x 4
  DESTINATION_AIRPORT media_atraso CITY      STATE
  <chr>                <dbl> <chr>    <chr>
1 ABE                  5.80 Allentown PA
2 ABI                  4.08 Abilene   TX
3 ABQ                  5.61 Albuquerque NM
4 ABR                 -3.39 Aberdeen  SD
5 ABY                  8.68 Albany    GA
6 ACK                  4.91 Nantucket MA
```

```
aeropertos_por_estado <- airports %>%
  group_by(STATE) %>%
  summarize(n_aeropertos = n()) %>%
  arrange(desc(n_aeropertos))

aeropertos_por_estado
```

4

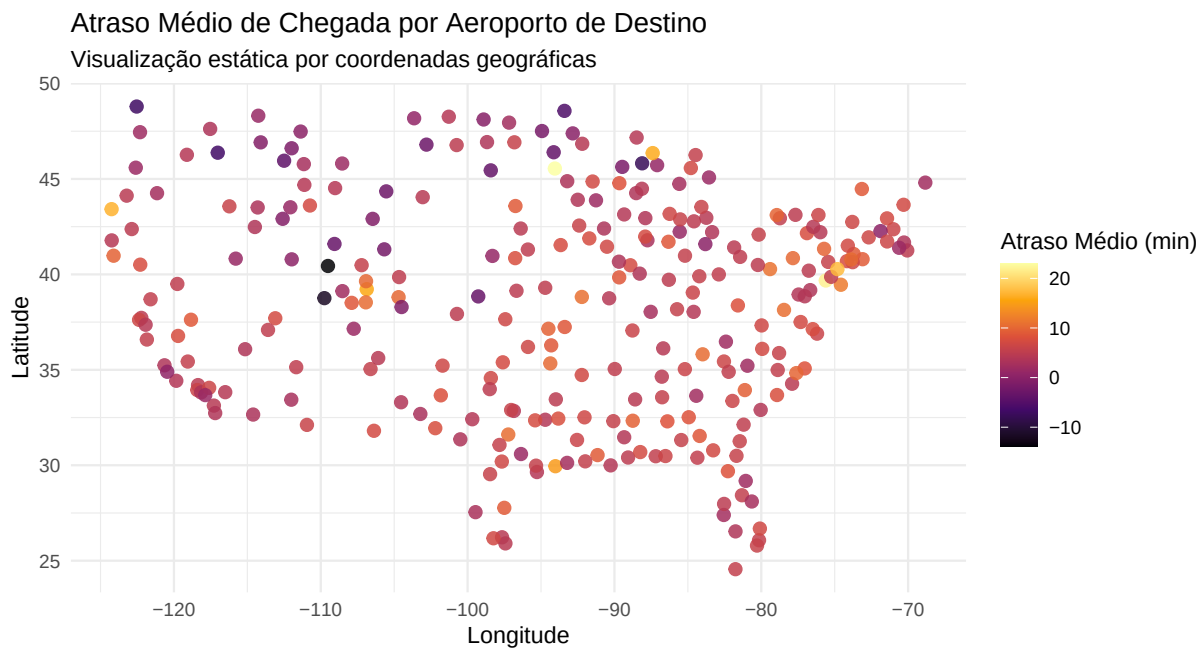
4

```

    "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, "<br>",
    "<b>Cidade:</b> ", CITY, " - ", STATE, "<br>",
    "<b>Atraso Médio:</b> ", round(media_atraso, 2), " min"
  )
)

# Plot estático para o PDF
dados_mapa %>%
  filter(LONGITUDE > -130 & LONGITUDE < -60 & LATITUDE > 20) %>%
  ggplot(aes(x = LONGITUDE, y = LATITUDE, color = media_atraso)) +
  geom_point(size = 2.5, alpha = 0.85) +
  scale_color_viridis_c(option = "inferno", name = "Atraso Médio (min)") +
  coord_fixed(1.3) +
  labs(
    title = "Atraso Médio de Chegada por Aeroporto de Destino",
    subtitle = "Visualização estática por coordenadas geográficas",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme_minimal()

```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 00:50:50 -03"
```